

# HW 6 Solutions

ECE 310: Digital Signals Processing, Spring 2025

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1

## Solution:

$$y[n] - y[n-1] = x[n]$$

a) Taking the z-transform of both sides:

$$Y(z) - Y(z)z^{-1} = X(z)$$

$$Y(z)(1 - z^{-1}) = X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \boxed{\frac{1}{1 - z^{-1}}, \text{ROC: } |z| > 1}$$

b) Taking the inverse z-transform of  $H(z)$  and using the fact that the system is causal:

$$\boxed{h[n] = u[n]}$$

c): The frequency response  $H_d(\omega)$  is not equal to  $H(z = e^{j\omega})$ . This can be readily seen by the ROC of  $H(z)$  being  $|z| > 1$ , barely not including the unit circle (defined as the set of all  $z$  such that  $|z| = 1$ ). As such,  $H(z)$  does not apply at values of  $z$  satisfying  $|z| = 1$ , meaning the two will not be equal. This can be readily seen by the fact that

$$H_d(\omega) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n} = \sum_{n=-\infty}^{\infty} u[n]e^{-j\omega n} = \sum_{n=0}^{\infty} e^{-j\omega n} \neq \frac{1}{1 - e^{-j\omega}} = H(z = e^{j\omega}).$$

**Grading:** 15 points

- (a)
  - +3 correct method setup
  - +2 correct  $H(z)$
  - +1 correct ROC
- +3 (b) correct impulse response
- +6 (c) reasonable explanation

2

## Solution:

(a) Take the DTFT of both sides and then apply the shifting property, we have:

$$Y_d(\omega) = X_d(\omega) (1 + e^{-j10\omega})$$

$$\frac{Y_d(\omega)}{X_d(\omega)} = e^{-5j\omega} (e^{j5\omega} + e^{-j5\omega})$$

$$H_d(\omega) = e^{-5j\omega} 2 \cos(5\omega)$$

So the magnitude

$$|H_d(\omega)| = 2|\cos(5\omega)|$$

and the Phase

$$\angle H_d(\omega) = \begin{cases} -5\omega & \text{if } \cos(5\omega) \geq 0 \\ -5\omega + \pi & \text{if } \cos(5\omega) < 0 \end{cases}$$

(b) Since the system is real ( $H_d(-\omega) = H_d(\omega)$ ) and the input is a purely sinusoidal signal, the output can be easily found as by using the system property to sinusoid inputs, given by:

$$x[n] = \cos(\omega_0 n + \phi) \rightarrow y[n] = |H_d(\omega_0)| \cos(\omega_0 n + \phi + \angle H_d(\omega_0))$$

(i) We can write:

$$y[n] = \left| H_d\left(\frac{\pi}{10}\right) \right| \cos\left(\frac{\pi}{10}n + \angle H_d\left(\frac{\pi}{10}\right)\right) + 3\left| H_d\left(\frac{\pi}{3}\right) \right| \sin\left(\frac{\pi}{3}n + \frac{\pi}{10} + \angle H_d\left(\frac{\pi}{3}\right)\right)$$

Now we know:

$$H_d\left(\frac{\pi}{10}\right) = 2 \cos\left(\frac{\pi}{2} e^{-j\frac{\pi}{2}}\right) = 0$$

$$H_d\left(\frac{\pi}{3}\right) = 2 \cos\left(\frac{5\pi}{3} e^{-j\frac{5\pi}{3}}\right) = e^{-j\frac{5\pi}{3}}$$

So the output is:

$$y[n] = 3 \sin\left(\frac{\pi n}{3} + \frac{\pi}{10} - \frac{5\pi}{3}\right)$$

(ii) Similar, we can write:

$$y[n] = 5|H_d(0)| \cos(\angle H_d(0)) + 3\left| H_d\left(\frac{2\pi}{5}\right) \right| \cos\left(\frac{2\pi n}{5} + \frac{\pi}{2} + \angle H_d\left(\frac{2\pi}{5}\right)\right)$$

Where:

$$H_d(0) = 2$$

$$H_d\left(\frac{2\pi}{5}\right) = 2 \cos(2\pi) e^{-j2\pi} = 2$$

So the final answer is:

$$y[n] = 10 + 6 \cos\left(\frac{2\pi}{5}n + \frac{\pi}{2}\right)$$

Plot is shown in Figure 1. Note that we keep the phase in the range  $[-\pi, \pi]$  by adding  $2k\pi$ .

**Grading:** 25 points

- +3 Magnitude response
- +3 Magnitude response plot
- +3 Phase response
- +3 Phase response plot
- +4 identifying all frequency components and +3 Correct response for b.i
- +4 identifying all frequency components and +2 Correct response for b.ii

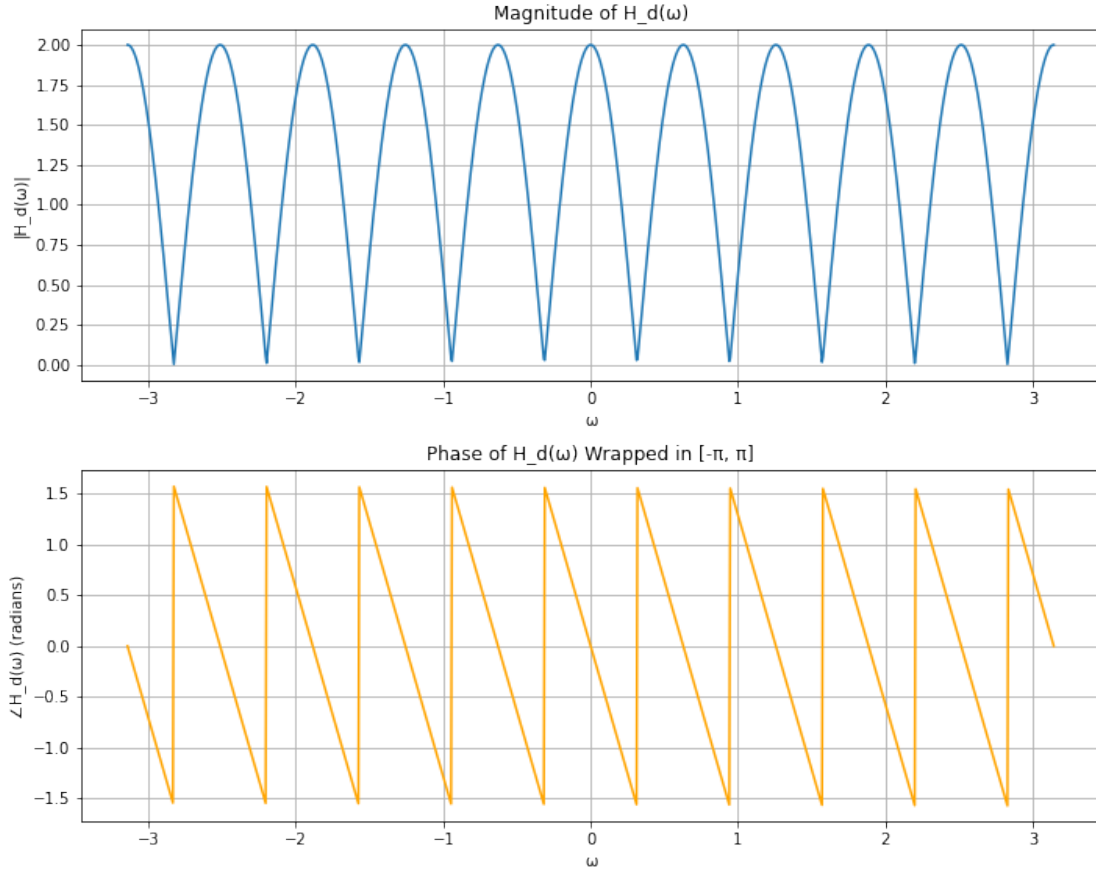


Figure 1: Problem 2 magnitude and phase plot

3

### Solution:

(a) According to the response of LTI systems, we have

$$y[n] = 2H_d(0) - 8H_d\left(\frac{\pi}{4}\right)e^{j(\frac{\pi}{4}n + \frac{\pi}{4})} + H_d\left(\frac{\pi}{2}\right)j^n$$

where  $j^n = (e^{j\pi/2})^n$ , so the third component has a frequency  $\pi/2$ .

We can use the given formula to show that  $H_d(0) = 0$ ,  $H_d(\frac{\pi}{4}) = \frac{\pi}{4}e^{j/\sqrt{2}}$ , and  $H_d(\frac{\pi}{2}) = \frac{\pi}{2}e^j$ , and therefore

$$y[n] = -2e^{j/\sqrt{2}}e^{j(\frac{\pi}{4}n + \frac{\pi}{4})} + \frac{\pi}{2}e^j j^n$$

(b) Since the system is not real-valued ( $H_d^*(\omega) \neq H_d(-\omega)$ ), we need to break  $\cos$  into exponentials, then apply the formula:

$$y[n] = 3H_d(0) + 5H_d\left(\frac{\pi}{4}\right)e^{j(\frac{\pi}{4}n + \frac{\pi}{4})} + 5H_d\left(-\frac{\pi}{4}\right)e^{-j(\frac{\pi}{4}n + \frac{\pi}{4})} - H_d\left(\frac{\pi}{2}\right)j^n$$

We can use the given formula to show that  $H_d(0) = 0$ ,  $H_d(\frac{\pi}{4}) = \frac{\pi}{4}e^{j/\sqrt{2}}$ ,  $H_d(-\frac{\pi}{4}) = -\frac{\pi}{4}e^{-j/\sqrt{2}}$ , and

$H_d(\frac{\pi}{2}) = \frac{\pi}{2}e^j$ , therefore

$$\begin{aligned}
 y[n] &= 5\frac{\pi}{4}e^{j/\sqrt{2}}e^{j(\frac{\pi}{4}n+\frac{\pi}{4})} - 5\frac{\pi}{4}e^{-j/\sqrt{2}}e^{-j(\frac{\pi}{4}n+\frac{\pi}{4})} - \frac{\pi}{2}e^j j^n \\
 &= 5\frac{\pi}{4}(e^{j/\sqrt{2}}e^{j(\frac{\pi}{4}n+\frac{\pi}{4})} - e^{-j/\sqrt{2}}e^{-j(\frac{\pi}{4}n+\frac{\pi}{4})}) - \frac{\pi}{2}e^j j^n \\
 &= 5\frac{\pi}{4}2j \sin(\frac{\pi}{4}n + \frac{\pi}{4} + \frac{1}{\sqrt{2}}) - \frac{\pi}{2}e^j j^n \\
 &= \boxed{5\frac{\pi}{2}j \sin(\frac{\pi}{4}n + \frac{\pi}{4} + \frac{1}{\sqrt{2}}) - \frac{\pi}{2}e^j j^n}
 \end{aligned}$$

**Grading:** 20 points

- 10 points each for (a) (b)
  - +6 for identifying all frequency components
  - +4 for correct result

## 4

### Solution:

We can find the frequency response  $H_d(\omega)$  by taking the DTFT of both sides:

$$Y_d(\omega) - \frac{1}{\sqrt{4}}e^{-j\omega}Y_d(\omega) = X_d(\omega) \implies H_d(\omega) = \frac{1}{1 - \frac{1}{\sqrt{4}}e^{-j\omega}}.$$

We can split  $x[n]$  into eigenfunctions as follows:

$$x[n] = 5e^{j0 \times n} + \frac{\cos(\frac{\pi}{4})}{j}e^{j\frac{\pi}{2}n} - \frac{\cos(\frac{\pi}{4})}{j}e^{-j\frac{\pi}{2}n} + 2e^{j\pi n}.$$

The eigenfunction property can be applied to show that

$$y[n] = 5H_d(0) + \frac{\cos(\frac{\pi}{4})}{j}H_d\left(\frac{\pi}{2}\right)e^{j\frac{\pi}{2}n} - \frac{\cos(\frac{\pi}{4})}{j}H_d\left(-\frac{\pi}{2}\right)e^{-j\frac{\pi}{2}n} + 2H_d(\pi)e^{j\pi n},$$

where  $H_d(0) = 1/(1 - \frac{1}{\sqrt{4}}) = 2$ ,  $H_d(\pi/2) = 1/(1 + \frac{1}{\sqrt{4}}j) = \frac{2}{\sqrt{5}}e^{-j \tan^{-1}(1/2)} = 0.8 - j0.4 = H_d^*(-\pi/2)$ , and  $H_d(\pi) = 1/(1 + \frac{1}{\sqrt{4}}) = 2/3$ . As such,

$$\begin{aligned}
 y[n] &= 10 + \frac{4}{3}e^{j\pi n} + \frac{\cos(\frac{\pi}{4})}{j} \left( \frac{1}{1 + \frac{1}{\sqrt{4}}j}e^{j\frac{\pi}{2}n} - \frac{1}{1 - \frac{1}{\sqrt{4}}j}e^{-j\frac{\pi}{2}n} \right) \\
 &= 10 + \frac{4}{3}e^{j\pi n} + \frac{\cos(\frac{\pi}{4})}{j} \frac{2}{\sqrt{5}} \left( e^{-j \tan^{-1}(1/2)}e^{j\frac{\pi}{2}n} - e^{j \tan^{-1}(1/2)}e^{-j\frac{\pi}{2}n} \right) \\
 &= 10 + \frac{4}{3}e^{j\pi n} + \frac{\cos(\frac{\pi}{4})}{j} \frac{2}{\sqrt{5}} \left( e^{j(\frac{\pi}{2}n - \tan^{-1}(1/2))} - e^{-j(\frac{\pi}{2}n - \tan^{-1}(1/2))} \right) \\
 &= 10 + \frac{4}{3}e^{j\pi n} + \cos\left(\frac{\pi}{4}\right) \frac{4}{\sqrt{5}} \sin\left(\frac{\pi}{2}n - \tan^{-1}\left(\frac{1}{2}\right)\right) \\
 &= \boxed{10 + \frac{4}{3}e^{j\pi n} + 2\sqrt{\frac{2}{5}} \sin\left(\frac{\pi}{2}n - \tan^{-1}\left(\frac{1}{2}\right)\right)}.
 \end{aligned}$$

**Grading:** 20 points

- +4 correct frequency response  $H_d(\omega)$
- +12 correctly applying eigenfunction property for each frequency component.
- +4 correct output  $y[n]$

5

### Solution:

We know that the system is real LSI, implying that it is fully described by a real-valued  $h[n]$  such that  $y[n] = h[n] * x[n]$ . This implies that it can be described with a conjugate symmetric frequency response,  $H_d(\omega) = H_d^*(-\omega)$ .

Given the relationship between  $x[n]$  and  $y[n]$  and the eigenfunction property, we can deduce that 3 maps to 9,  $\cos(\frac{\pi}{4}n + 10^\circ)$  maps to  $2 \sin(\frac{\pi}{4}n + 10^\circ) = 2 \cos(\frac{\pi}{4}n - 80^\circ)$  and  $\sin(\frac{\pi}{3}n + 25^\circ)$  maps to 0. As such, we can write that  $9 = 3H_d(0)$ , or  $H_d(0) = 3$ . We can also deduce that  $H_d(\pi/3) = 0$  from the latter mapping. Finally, we can write

$$\begin{aligned} \cos\left(\frac{\pi}{4}n + 10^\circ\right) &= \frac{1}{2}e^{j(\frac{\pi}{4}n + 10^\circ)} + \frac{1}{2}e^{-j(\frac{\pi}{4}n + 10^\circ)} \rightarrow \\ H_d\left(\frac{\pi}{4}\right) \frac{1}{2}e^{j(\frac{\pi}{4}n + 10^\circ)} + H_d^*\left(\frac{\pi}{4}\right) \frac{1}{2}e^{-j(\frac{\pi}{4}n + 10^\circ)} &= e^{j(\frac{\pi}{4}n - 80^\circ)} + e^{-j(\frac{\pi}{4}n - 80^\circ)} \end{aligned}$$

From this, we can deduce that  $H_d\left(\frac{\pi}{4}\right) = 2e^{-j90^\circ} = -2j$ .

We can finally find the response  $\tilde{y}[n]$  to input  $\tilde{x}[n]$  via

$$\tilde{y}[n] = 5H_d(0) + 2H_d\left(\frac{\pi}{4}\right) \cos\left(\frac{\pi}{4}n - 75^\circ\right) + 10H_d\left(-\frac{\pi}{3}\right) \cos\left(-\frac{\pi}{3}n + 25^\circ\right).$$

By conjugate symmetry, since  $H_d(\pi/3) = 0$ ,  $H_d(-\pi/3) = H_d(\pi/3)^* = 0$ . The other two coefficients remain the same from above. As such, we can express

$$\tilde{y}[n] = 15 - 4 \cos\left(\frac{\pi}{4}n + 15^\circ\right).$$

**Grading:** 20 points

- + 16 correctly using eigenfunction property to find  $H_d(0), H_d(\frac{\pi}{4}), H_d(\frac{\pi}{3}), H_d(-\frac{\pi}{3})$
- + 4 correct response  $\tilde{y}[n]$